Skin Segmentation Using YCBCR and RGB Colour Models

Koju, R., & Joshi, S. R.,
Comparative analysis of color image watermarking technique in RGB, YUV, and YCbCr color channels, Nepal Journal of Science and Technology, vol. 15, no. 2, pp. 133–140, 2014.

https://www.researchgate.net/profile/Jyoti-Waykule/publication/290440563_Skin_Segmentation_Using_YCBCR_and_RGB_Color_Models/links/5698d18d08ae1c4279067661/Skin-Segmentation-Using-YCBCR-and-RGB-Color-Models.pdf

Summary

This paper looks at how skin segmentation can be done using RGB and YCbCr colour models, which is an important step for tasks like face detection. It compares both approaches and shows how YCbCr works better because it separates brightness (luminance) from colour (chrominance), making it easier to isolate skin tones. The method uses chrominance thresholding and statistical modelling to detect skin pixels across different lighting conditions and skin types. While both RGB and YCbCr can be used for segmentation, YCbCr gives more consistent and accurate results. The study concludes that separating luminance from chrominance improves detection reliability in real-world image processing applications.

Relevance

This paper supports my skin detection section by explaining why YCbCr is more effective than RGB for identifying skin regions. It directly relates to my implementation using Cb and Cr thresholding in the code walkthrough.